

COPING WITH ENVIRONMENTAL UNCERTAINTY AND BOUNDARY SPANNING IN THE PRODUCT MANAGER'S ROLE

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Evidence suggests that product managers' need to communicate across organizational and environmental boundaries under conditions of uncertainty can give rise to powerful role pressures of conflict and ambiguity. These pressures are generally associated with negative or dysfunctional personal outcomes such as job-related tension, dissatisfaction, and poorer performance. Moreover, in situations where role conflict is particularly high, experienced product managers are susceptible to "burnout." Some practical approaches to overcoming these difficulties are discussed.

The system of employing product managers to oversee all aspects of the marketing of individual products was first adopted by Procter and Gamble more than half a century ago. Since then it has become generally accepted by the consumer packaged goods industry and is currently practiced by more than 85 percent of industry participants. The product manager's

**Note: The research for this article was done among product managers of consumer goods companies. However, the editor feels that its information is of such significant importance, that it is being printed in this journal.*

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task is to harness and integrate both internal and external resources to support his or her assigned product or brand. Therefore, a major part of the job entails communication and interaction with other departments of the organization and with numerous external sources. This activity of going outside the boundary of the marketing department and interfacing with both corporate departments and outside contacts is often referred to as *boundary spanning*. (See Figure 1).

Boundary Spanning and Role Pressures

According to one description offered, the product manager "acts as a focal point for screening all product improvement ideas that flow from the consumer, the trade, and the field selling organization." Therefore they must cope with the diverse expectations and demands of a variety of internal and external constituencies, such as production departments and advertising agencies. Since the success of the product is also of concern to these other groups, each has vested interests in influencing or controlling the product manager's behavior. One product manager remarked, "I feel like a real politician at times—I'm always lobbying for support from

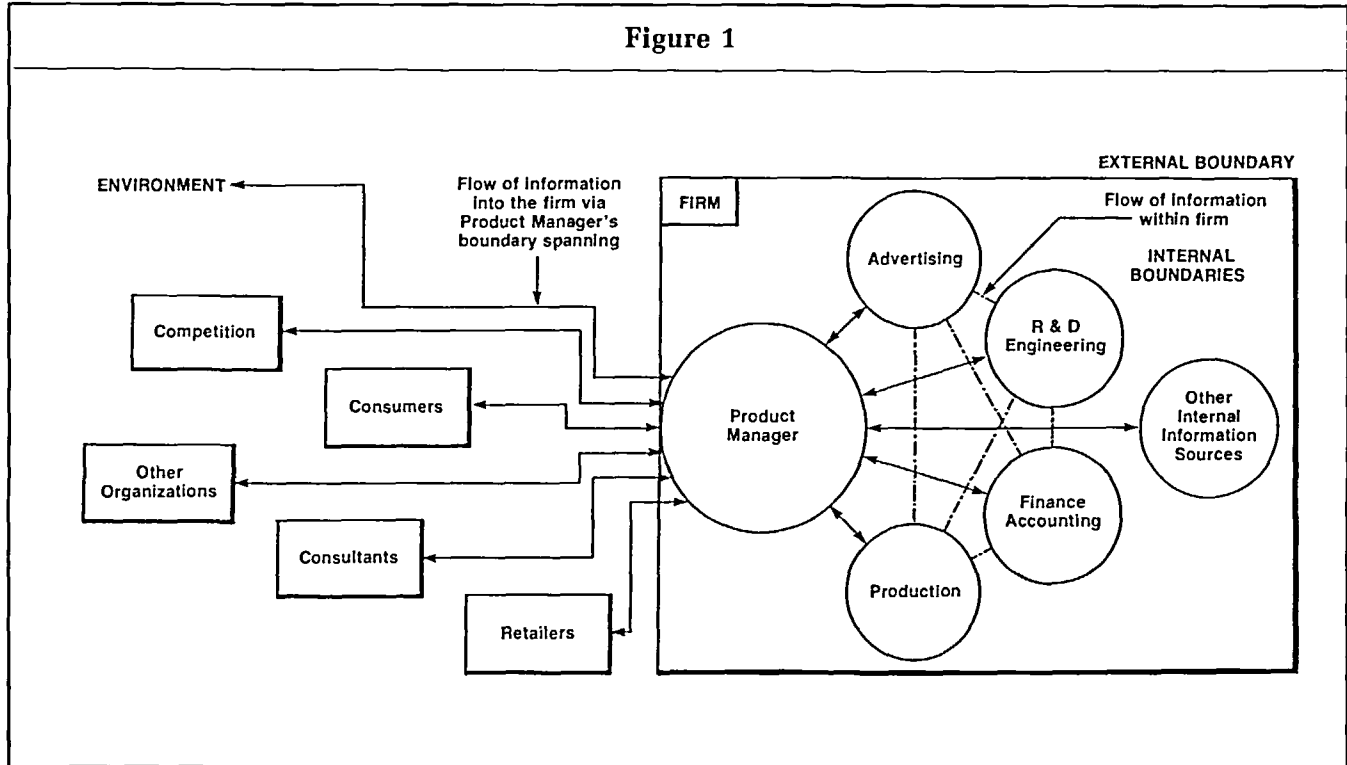
other departments."

The product manager continually has to adapt the marketing mix of his product line to a changing market environment.

In addition, the product manager continually has to adapt the marketing mix of his product line to a changing market environment. Since each adaptive change in the product or marketing mix is of interest to various groups, it seems likely that dynamic environmental conditions might have an impact on the frequency and importance of boundary spanning activities by product managers. The managers interviewed in this study often expressed feelings of tension and of being "caught in the middle." One interviewed at Gillette Safety Razor Division commented, "I feel like I'm in a pressure cooker at times, trying to juggle all the different departments' expectations."

These comments suggest that boundary spanning and environmental uncertainty together give rise to role pressures acting on product managers. These pressures involve role conflicts (i.e., attempts to comply with conflicting demands)

Figure 1



and role ambiguities (i.e., uncertainty about goals or responsibilities). Figure 2 illustrates the interplay of influences and expectations that act as potential stressors on product managers.

Model and Research Questions of the Study

This study investigates several behavioral variables relating to the boundary spanning role of product managers. The basic assumption is that a high degree of environmental uncertainty results in an increase in boundary spanning activity and that together these factors act to increase the role pressures (conflict and ambiguity) experienced by the product manager. The model (see Figure 3) also predicts that higher levels of role conflict and role ambiguity, in turn, lead to lower job satisfaction, higher tension, and a poorer perceived performance. It is also expected that individual personality differences might serve to moderate the connection between role pressures and these personal outcomes. Details of each hypothesis and their rationales are as follows:

Research Hypothesis 1. *The greater the environmental uncertainty, the greater the degree of boundary spanning activity by the product manager.*

Faced with increased environmental turbulence, the product manager would seem to require additional market and product information, and opinions from different perspectives of various constituencies. A stable market, by contrast, diminishes the need for boundary spanning activity to collect more information.

Research Hypothesis 2. *The greater the environmental uncertainty, the higher the levels of role conflict and role ambiguity experienced by the product manager.*

It is difficult to resolve or avoid conflicts of interest between groups when they are each attempting to adapt to environmental turbulence. In these conditions, goals or objectives may also be subject to revision, thereby increasing the role ambiguity experienced by the product manager.

Research Hypothesis 3. *The greater the amount of boundary spanning activity by the product manager, the higher the level of role conflict and role ambiguity.*

Boundary spanning activities are directly associated with role conflict and ambiguity because of the need to accommodate diverse expectations and goals, respectively.

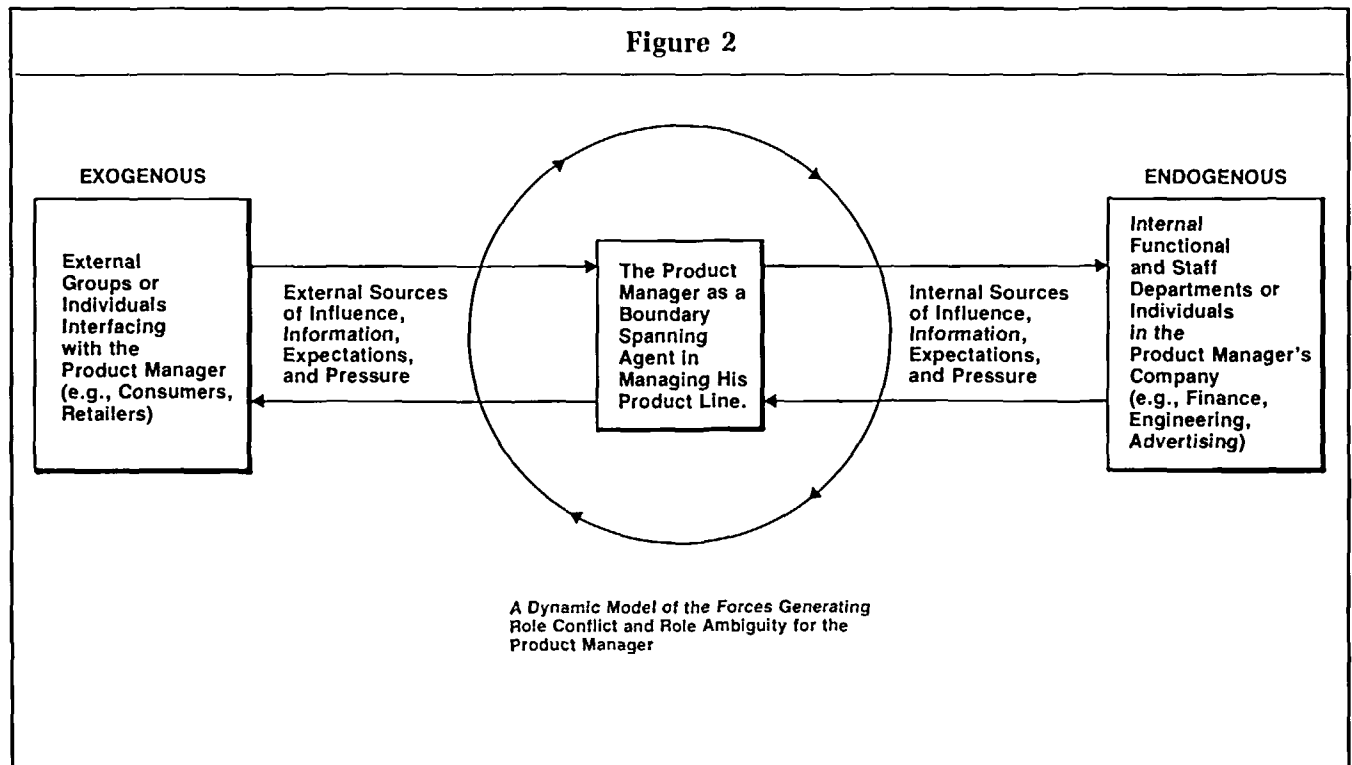
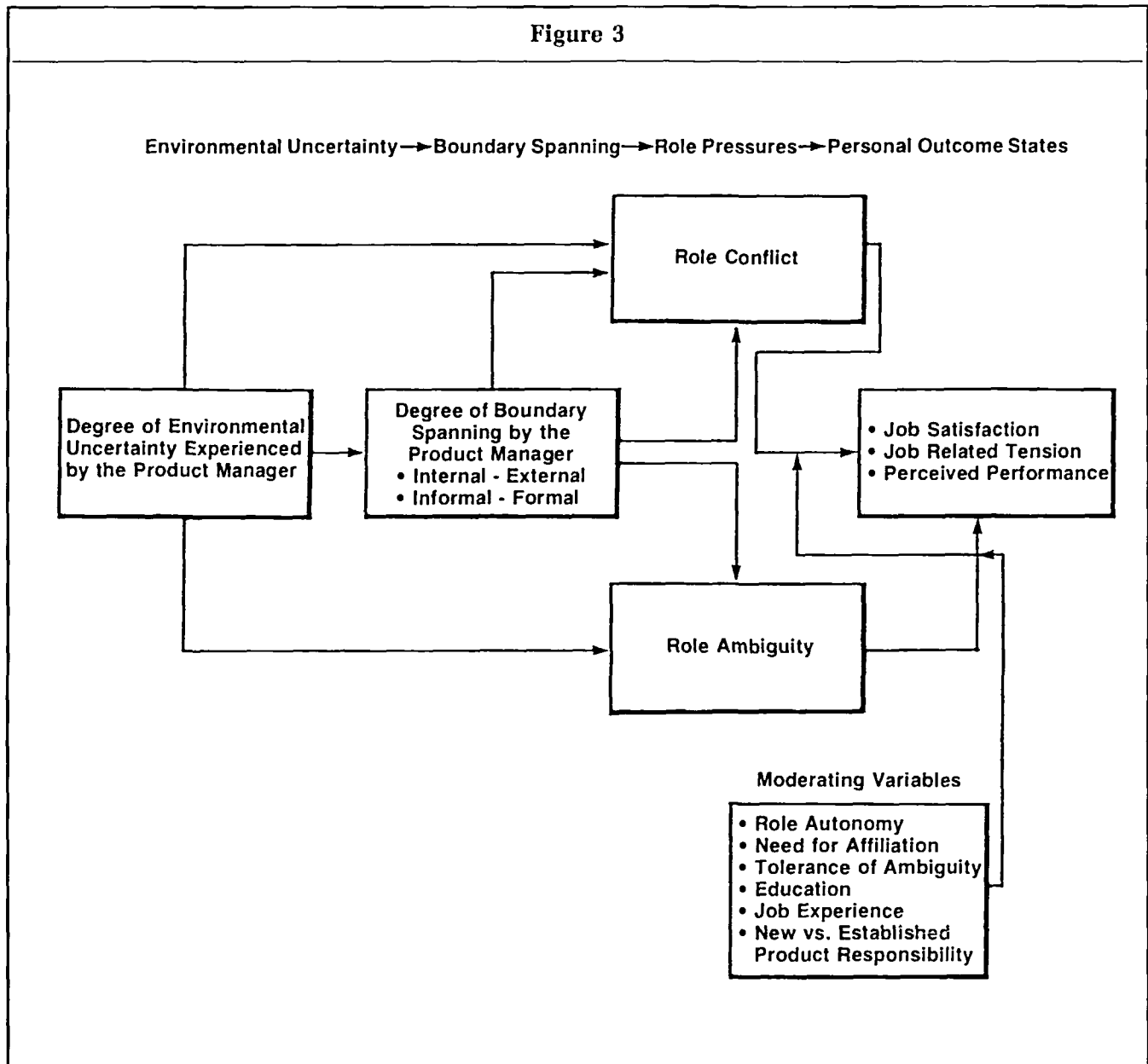


Figure 3



Research Hypothesis 4. Role conflict and role ambiguity are both associated with (a) greater job tension, (b) lower job satisfaction, and (c) lower perceived performance.

Ambiguity of goals and incompatibility of expectations are potential stressors that may result in increased job tension. Once this tension exceeds moderate levels, reductions in job satisfaction and performance may be expected. Several earlier studies have established general relationships of this sort between stressors, task difficulty, and task performance.

Research Hypothesis 5. Personality and situa-

tional variables moderate the above relationships between role pressures and personal outcomes.

It is difficult to resolve or avoid conflicts of interest between groups when they are each attempting to adapt to environmental turbulence.

Personality variables such as need-for-affiliation and tolerance-of-ambiguity, as well as education and experience, have been shown to mediate

reactions to conflict and ambiguity in some situations. Accordingly, these are expected to act as moderators of the product manager's responses to role pressures. In particular, since social and formal contacts or interactions are such an integral part of boundary spanning activity, the need for affiliation was expected to lessen the adverse personal outcomes associated with high levels of role pressure. Tolerance of ambiguity measures the ability to respond positively to ambiguous situations or partially defined problems. The concept was originally developed from the psychoanalytic tradition by psychologist Frenkel-Brunswick in the early postwar period.¹³ In the present study, this personality trait was expected to moderate the effects of role ambiguity. In addition to these personality measures, the job related variables of role autonomy and "new versus established product responsibility" were also expected to influence the personal reactions experienced by the product manager.

Study Description and Measures Used

Questionnaires were mailed to 449 product managers in consumer packaged goods industries, drawn from the Fortune 500.⁶ The sampled industries included food, proprietary drugs and cosmetics, soaps, cleansers and allied products, soft drinks, tobacco, paper, and liquor. All of these industries have been using product managers for many years. The final usable sample was 170 completed questionnaires. All of the items in the composite questionnaire instrument were validated by prior research. (See Appendix 1.)

Perceived Environmental Uncertainty

Nine items measured perceived uncertainty in the marketplace. Each item examined a different aspect of environmental uncertainty. For example: "How often do you have the necessary information on competitors? How often do changes in social, economic, political, or technological conditions outside your company affect your product-related decision making?"

Boundary Spanning Activity

Eleven items were used to measure the extent of boundary spanning by the product managers. Total boundary spanning activity was separated into formal versus informal and internal versus external communications (excluding strictly social contacts). Separate items were used for verbal

and written communications. A typical item was: "How many times a week do you send or receive formal written communication, in the form of reports or data from people (a) outside the marketing department but within your company? (b) outside your company?"

Role Pressures

Role conflict is defined as incompatibility of expectations concerning the product manager's role. Ten Likert-scale items were used to measure role conflict along various dimensions such as intrasender (arising from the same source), and intersender conflict. For example: "I work with two or more groups who operate quite differently." "I receive incompatible requests from two or more people." Role ambiguity is defined as inconsistencies about job responsibilities and evaluation-related goals. Six items were used, such as "Clear, planned goals and objectives exist for my job" and "I know exactly what is expected of me."

Personal Outcome States

Job satisfaction was measured by five items, such as "In light of my career expectations I am satisfied with my present position." Eleven other items concerned job tension; for example, "My job tends to directly affect my health" and "I breathe a sigh of relief when I am able to travel out of town." The third personal outcome state variable, perceived performance, was measured using two items; for example, "How do you rate yourself in terms of managing your product line?" On this point it should be noted that previous research¹⁷ has found that self-ratings of performance correlate strongly with more objective measures, and may actually be tougher and more accurate than ratings by others.

Personality and Situational Moderators

The situational variable of role autonomy was measured through six Likert-scaled items,⁹ such as, "I have enough freedom as to how I do my work." Tolerance of ambiguity was measured by six items from Budner's scales,¹ such as "An expert who doesn't come up with a definite answer probably doesn't know too much."

Need for affiliation was measured by seven Likert-scaled items,⁴ such as "Often I attend social gatherings just to be with others." Educational background and job experience (with both

company and product) were also measured.

Findings of the Study

Analysis of the questionnaire data revealed the following:

1. Environmental uncertainty has only a slight effect on the amount of boundary spanning activity by product managers (Appendix 2).
2. The greater the perceived uncertainty in the environment, the greater the role pressures experienced by product managers (Appendix 3).
3. Boundary spanning activity produces high levels of role conflict, but it does not directly affect the role ambiguity experienced by product managers. (Appendix 4).
4. Intense role pressures lead to increased levels of job-related tension, reduced job satisfaction, and lower perceived performance (Appendix 5).

Moderators

The product managers' levels of experience and educational background were found to be the principal moderators of these dysfunctional effects of role pressures (Appendix 6).

- 5a. The level of job related tension arising from role pressures tended to increase with on-the-job experience as a product manager (or with the present product).
- 5b. Educational level can help to reduce job-related tension and dissatisfaction in some situations.
- 5c. High need-for-affiliation scores were associated with better perceived performance, but only where role pressures were high.
- 5d. High tolerance-of-ambiguity scores did not indicate any special ability to cope with role pressures.

Discussion

Determinants of Boundary Spanning

Two of the findings regarding uncertainty and boundary spanning were surprising in relation to the hypothesized model (Figure 3). First, environmental uncertainty does not appear to

have much influence on the amount of boundary spanning activity actually carried out by product managers. There are several plausible explanations for this weak link in the model. Individual product managers may have a usual propensity to engage in boundary spanning by communicating with other departments and with external sources. One group product manager stated that "if a product manager spends more than half of the day in his office, he's not collecting enough information about the product." This comment suggests that boundary spanning is constrained more by the product manager's general orientation to the job than by the particular environmental conditions. Other studies have reported that product managers frequently do not, in fact, have sufficient external sources of information, suggesting another constraint on the frequency of boundary spanning.

Role Ambiguity or Role Conflict

A second surprise is that role ambiguity does not arise simply because of boundary spanning activity. Role ambiguity is more a function of environmental uncertainty. This finding suggests that the dysfunctional effects of boundary spanning activities can be traced specifically to the role conflict involved.

Role conflict is defined as incompatibility of expectations concerning the product manager's role.

In addition, role conflict is also strongly associated with environmental uncertainty, such that uncertainty intensifies the conflicts arising from a relatively unchanging amount of boundary spanning activity. The effects of this conflict and ambiguity were overwhelmingly negative in terms of all personal outcome scores for the product managers. We can now say that high levels of boundary spanning activity lead to dysfunctional increases in job-related tension and dissatisfaction, and this is particularly due to the associated role conflict. Moreover, environmental uncertainty tends to intensify the dysfunctional effects. It seems that product managers must be on guard against high levels of tension resulting indirectly from uncertainty in the environment and directly from role conflict and ambiguity.

Mitigating the Dysfunctional Effects of Role Pressures

Several other studies have shown that moderate tension may have a positive motivational effect, while high levels of tension or stress may have negative effects, making the individual dysfunctional in job performance. In this study, role conflict and ambiguity are both strongly associated with tension, dissatisfaction, and lower performance for product managers. In some cases, these negative outcomes might induce the individual to adopt coping strategies such as withdrawal from conflict through absenteeism or even alcoholism and drug abuse, personal responses that ultimately intensify the personal and organizational difficulties. Stress has been shown to be a major cause of low productivity, absenteeism, poor decisions, bad judgment, misallocation of resources and bad morale.² Therefore senior managers and personnel staff should consider various methods for mitigating the harmful effects of the role pressures acting on product managers.

It is difficult to reduce the level of environmental uncertainty that the product manager faces or the amount of boundary spanning activity carried out. Therefore, to minimize deleterious outcomes, the emphasis must be placed on managing the consequences of these forces rather than on trying to eliminate factors that produce role pressures. Role conflict, particularly inter-sender conflict, might be reduced by the development of clear procedures to resolve these conflicts as they arise. Also, the product manager should be kept informed of any changes in company policy, together with the reasons for the changes, so that conflicting expectations can be evaluated in relation to relevant aspects of policy. Indeed, several of the interviewed product managers at firms such as Gillette suggested that more direct exposure to overall company objectives would be helpful. For example, the vice-president of marketing should acknowledge company objectives and make them known to product managers. Setting or clarifying brand objectives might also help to solve the problem of role ambiguity. According to the senior managers interviewed, an MBO approach involving agreed-upon goals could be helpful here. The development of a clearly written job description specifying responsibilities and authority will help minimize the product manager's stress. A different approach to the management of role

pressures would be to introduce stress audits. When personal stress exceeds acceptable levels, senior management could be alerted. A product management audit system could also help to identify unusual pressures upon product managers.⁸

Recruitment

A related approach concerns the creation of a better fit between the individual and the role. Given the inevitable levels of conflict and ambiguity associated with product management, particularly where environmental uncertainty is high, the challenge is to identify suitable individuals as product managers. Frito-Lay, for example, looks for individuals with maturity, communication skills, common sense, and interpersonal competencies. General Foods chooses candidates who have good observation skills and an understanding of human nature. Candidates should be invited to seriously consider the fit between the exacting demands of the product manager's role and their own strengths and weaknesses.

Analysis of the moderator variables in the present study carries several messages for this recruitment and training process. It appears that those with a higher level of formal education may feel more comfortable operating in the absence of clearly defined goals and may experience less dissatisfaction under conditions of intense role conflict. The personality variable "need for affiliation" may be of some relevance to the recruitment process. This variable was associated with higher perceived performance in conditions of intense role conflict and high ambiguity. (However, further research on this point is advisable, since the same trait was also associated with higher tension under conditions of high ambiguity.)

A further finding of the study has potentially important implications for the training and retention of product managers. Although experience was generally related to higher perceived performance, it was strongly associated with *increased* levels of tension under conditions of intense conflict and high ambiguity. Evidently, product managers do not learn to cope with situations of intense role conflict; they simply "burn out." This burnout may be due to sheer exhaustion, but an alternative explanation is that the negative effects of intense role conflict are heightened rather than dampened as the manager's interpersonal relationships become

more developed over time. It is likely that product managers with Type A personalities are more stress prone and therefore more likely to become overloaded by stress.²

Summary and Conclusion

A number of interesting findings about the product manager's role have emerged from this study. First, the actual extent of boundary spanning activity seems to depend more on the individual product manager's approach to the job rather than on the degree of uncertainty in the market environment. Second, boundary spanning activity and environmental uncertainty act together in amplifying the role pressures acting on the product manager, so that these

may reach dysfunctional levels. The result is tension, dissatisfaction, and lower performance. Moreover, where role pressures are intense, experience in this job actually appears to increase the levels of tension—indicating a high risk of burnout where such conditions prevail. This finding should be of some concern to personnel staff, senior management, and, of course, to product managers themselves. Individuals differ in the extent to which they might fall victim to these stressors, yet environmental uncertainty and boundary spanning activity are unavoidable, and concomitant features of the product manager's world. Therefore, the careful recruitment and training of suitable personnel for the product management function is very much in the interests of the firm, as well as the individuals concerned.

Appendix

Appendix 1				
Scales, Items per Scale, Means, Standard Deviations, and Reliability Coefficients				
<u>Scale</u>	<u>Number of Items</u>	<u>Mean</u>	<u>Standard Deviation</u>	<u>Reliability Coefficient</u>
Environmental uncertainty				
perceived	9	41.1	5.9	.67
Boundary spanning	11 ^a	84.4	45.3	.74
internal	4	57.7	35.2	.71
external	4	19.9	15.8	.64
informal	4	43.6	29.3	.73
formal	4	23.9	16.8	.77
others	3	15.1	11.8	.52
Role perceptions				
role conflict	10	38.6	8.5	.84
intrasender	2	7.2	2.6	
intersender	4	14.9	3.8	
role ambiguity	6	15.6	4.7	.83
Personal outcome states				
job satisfaction	5	22.2	4.4	.75
job-related tension	11	36.3	9.6	.86
perceived performance	2	14.96	2.1	.54
Moderating variables				
role autonomy	6	28.4	4.9	.80
tolerance of ambiguity	6	14.4	3.5	.60
need for affiliation	7	27.6	3.9	.52

^aThis does not add to 19 since some of the subscales overlap.

Appendix 2**Relationships Between Environmental Uncertainty And Measures Of Boundary Spanning**

<u>Measure of Boundary Spanning</u>	<u>Perceived Environmental Uncertainty</u>
Aggregate boundary spanning	.11 ^a
Informal boundary spanning	.10 ^a
formal boundary spanning	.10 ^a
internal organizational boundary spanning	.10 ^a
external organizational boundary spanning	.03

a $\leq .10$.

The table reports the correlations between perceived environmental uncertainty and the components of boundary spanning activity. Most reached significance at the .01 level. The aggregate measure of boundary spanning by product managers was weakly but significantly correlated with perceived environmental uncertainty. Hence, it appears that uncertainty provides only a weak stimulus to increased boundary spanning activity by product managers.

Appendix 3**Environmental Uncertainty And Role Pressures**

<u>Role Pressures</u>	<u>Perceived Environmental Uncertainty</u>
Role conflict	.29 ^a
intersender conflict	.25 ^a
intrasender conflict	.17 ^b
Role ambiguity	.23 ^a

^a $p \leq 0.001$.

^b $p \leq 0.01$.

The table reports strong correlations between perceived uncertainty and both components of role conflict and role ambiguity. Therefore uncertainty amplifies the role pressures experienced by product managers.

Appendix 4**Relationships Between Measures Of Boundary Spanning, Role Conflict And Role Ambiguity**

<u>Measures of Boundary Spanning</u>	<u>Role Conflict</u>	<u>Role Ambiguity</u>
Aggregate boundary spanning	.28 ^a	.00
informal boundary spanning	.14 ^b	.10
formal boundary spanning	.24	.05
internal organizational boundary spanning	.22 ^a	.13 ^b
external organizational boundary spanning	.12 ^b	-.04

^a $p \leq 0.001$.

^b $p \leq 0.05$.

The results in the table show that although the measures of role conflict were strongly correlated with boundary spanning activity, role ambiguity was uncorrelated with boundary spanning.

Appendix 5
Relationships Between Role Conflict And Role Ambiguity And Job Satisfaction

<u>Role Perceptions</u>	<u>Job Satisfaction</u>	<u>Job-Related Tension</u>	<u>Perceived Performance</u>
Role conflict	-.24 ^a	.28 ^a	-.18 ^b
intersender conflict	-.20 ^a	.24 ^a	-.17 ^c
intrasender conflict	-.24 ^a	.24 ^a	-.17 ^c
Role ambiguity	-.28 ^a	.15 ^c	-.34 ^a

^a_p ≤ 0.001.

^b_p ≤ 0.01.

^c_p ≤ 0.05.

Role conflict and role ambiguity were strongly associated with lowered job satisfaction and heightened job-related tension. These role pressures were also correlated with lower perceived performance.

Appendix 6
Summary Table Of The ANOVA Tests On The Moderating Variables

	<u>F Statistic for Interaction Effect</u>
Role Conflict—Satisfaction Relationship	
Role autonomy	.144
Need for affiliation	.63
Tolerance of ambiguity	.004
Education	6.04 ^a (increased satisfaction)
Experience as a product manager	.25
Experience with present company as a product manager	.72
Experience with present product as a product manager	.15
New versus established product responsibility	.08
Role Conflict—Job-Related Tension Relationship	
Role autonomy	.92
Need for affiliation	.50
Tolerance of ambiguity	4.74 ^b (decreased tension)
Education	.18
Experience as a product manager	7.42 ^c (increased tension)
Experience with present company as a product manager	.72
Experience with present product as a product manager	6.50 (increased tension)
New versus established product responsibility	.39

Continued on next page

Appendix 6 continued
Summary Table Of The ANOVA Tests On The Moderating Variables

	<u>F Statistic for Interaction Effect</u>
Role Conflict—Perceived Performance Relationship	
Role autonomy	.06
Need for affiliation	7.02 ^c (increased performance)
Tolerance of ambiguity	.41
Education	.05
Experience as a product manager	.22
Experience with present company as a product manager	1.01
Experience with present product as a product manager	.37
New versus established product responsibility	.04
Real Ambiguity—Satisfaction Relationship	
Role autonomy	9.38 ^c (increased satisfaction)
Need for affiliation	.27
Tolerance of ambiguity	.13
Education	.24
Experience as a product manager	.42
Experience with present company as a product manager	1.06
Experience with present product as a product manager	.04
New versus established product responsibility	.32
Role Ambiguity—Job-Related Tension Relationship	
Role autonomy	.28
Need for affiliation	5.83 ^a (increased tension)
Tolerance of ambiguity	2.60
Education	7.25 ^c (lowered tension)
Experience as a product manager	.05
Experience with present company as a product manager	.41
Experience with present product as a product manager	6.20 ^c (increased tension)
New versus established product responsibility	.89
Role Ambiguity—Perceived Performance Relationship	
Role autonomy	.31
Need for affiliation	3.83 ^b (increased performance)
Tolerance of ambiguity	.02
Education	.03
Experience as a product manager	4.85 ^b (increased performance)
Experience with present company as a product manager	.15
Experience with present product as a product manager	.34
New versus established product responsibility	.26

^a $p \leq 0.01$.

^b $p \leq .05$.

^c $p \leq .001$.

END NOTES

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